SAM MAHDAVIAN, PhD, PMP, PMI-PBA

AI Solutions & Delivery Lead · Founder & CEO, Brane

Agentic systems that solve the business problem. Tested, governed, and safe in regulated production.

Orlando, FL · 407-808-3580 · sam@sammahdavian.ai · sammahdavian.ai · LinkedIn · GitHub · Google Scholar · AWS Blog

PROFESSIONAL SUMMARY

AI solutions leader who turns ambiguous business problems into agentic systems that hold up in regulated production. I work as a single-threaded owner from discovery and pre-sales through reference architecture, build, and adoption, and every system I ship has to clear one test: it solves the business problem, it is evaluated against real cases, and it is safe to run. Across healthcare, financial services, transportation, retail, and manufacturing I have shipped 40+ systems and 150+ agents that together produced $40M+ in revenue and $500M+ in identified client savings, and I have authored 50+ Claude skills that specialize those agents into repeatable, gated engines. I also co-founded and run Brane, an AI siting-and-planning startup, as CEO. For 15+ years I have been the translator between the C-suite that wants ROI and the engineers who build, pairing LLM/RAG and AWS engineering with executive-facing delivery. PhD (ML / applied AI); PMP; PMI-PBA.

CORE COMPETENCIES

Solutions & Pre-Sales: enterprise AI delivery, solution architecture, discovery to design to deploy to adopt, requirements elicitation across sectors, pre-sales and POCs, buyer-persona and commercial market analysis, ROI and business-case framing, single-threaded ownership, executive enablement.

Agentic AI & GenAI: LangChain/LangGraph, Strands SDK, CrewAI, Claude Code and custom Python; multi-agent orchestration; agent specialization through 50+ authored skills (workflow, capability, domain) with approval gates and append-only decision logs as agent memory; MCP tool design; RAG and vector DBs; guardrails, evals, and human-in-the-loop; prompt engineering; latency/cost/accuracy optimization; AWS Bedrock and AgentCore; Claude and OpenAI.

Production & MLOps: AWS (Bedrock, SageMaker, Lambda, Step Functions, S3, ECS/EKS), Docker, Kubernetes, CI/CD, IaC (Terraform/CDK), model serving and LLMOps, observability, model-risk governance, IAM and cloud security, HITRUST/HIPAA, FinOps.

Leadership & Delivery: team building and mentorship (orgs to ~25), roadmap and solution ownership, GTM, pricing and packaging, Agile/Scrum and SAFe, storytelling with data, hiring and talent development.

PROFESSIONAL EXPERIENCE

AArete LLC Director, Data Science & AI — Orlando, FL · 2022 - Present

- Designed the architecture and led the build of Doczy.ai, a multi-tenant GenAI platform on AWS (retrieval, prompt orchestration, agentic tool-calling, dynamic model selection, governance, observability). Scaled it to 2.5M+ contracts and 442B+ tokens at 99%+ accuracy under HITRUST/HIPAA, generating $15M+ in revenue and $330M+ in client savings. Architecture featured on the AWS Architecture Blog; issued a U.S. patent in 2025.
- Built 40+ systems and 150+ agents that each solve a specific business problem, most now in production, on LangChain/LangGraph, Strands SDK, Claude Code, and custom Python over AWS Bedrock and AgentCore. Across the portfolio these systems produced $40M+ in revenue and $500M+ in identified client savings. Every one is evaluated against real cases before release, tuned for latency, cost, and accuracy, and built human-in-the-loop and audit-first.
- Authored 50+ Claude skills across 12 engine repositories that turn one-off agent work into repeatable, gated procedures: workflow skills (investigate, requirements with an approval gate, build, test), capability skills (PHI/PII pre-flight, engine repeatability, cost observability), and domain skills per engine. Result: any teammate regenerates a deliverable from a clean clone with one command, and a strict split between LLM judgment and deterministic precision is enforced by the tooling, not by discipline.
- Built an AI claims-testing engine for a Medicaid managed-care plan that reads any provider contract and the payer's own configuration, proves what each claim should pay with no LLM in the pricing chain, and generates submittable test claims with verbatim receipts: 37 pipeline steps behind one command, 90 automated gates, 1,647 tests, three independent verification wheels, 28 packaged skills, and an adversarial reader stage that must clear every cut. Test authoring roughly 100× faster than by hand and repeatable across contracts and providers; progressed from business development to a signed SOW.
- Run discovery and pre-sales as a single-threaded owner across 10+ Fortune 500 engagements. I sit with executives and business teams, map the real pain by sector, and convert ambiguous problems into structured SOWs and roadmaps. Recently won a $700K+ engagement with an equipment-finance lender, plus $2.5M+ in competitive government and industry RFPs.
- Own production deployment (Docker, ECS/EKS, CI/CD, IaC, Bedrock/SageMaker serving) and authored the firm's AI-governance framework: model-risk management, bias and drift monitoring, explainability, prompt-injection defense, confidence-scored human-in-the-loop, and a four-stage maturity model that gates every system before it ships.
- Lead and mentor ~15 AI scientists and engineers across backend, frontend, and DevOps, plus a business-translation layer that turns stakeholder needs into specs and prompts. Run the firm-wide AI delivery program: ~$1.3M budget, 10+ concurrent workstreams, a six-gate value framework, and value-based pricing that protects margin as delivery cost falls.
- Lead executive AI-enablement workshops; one for 40+ leaders at an S&P 500 global manufacturer produced a 30% productivity increase and 20+ prioritized use cases. Earlier (Manager / Sr. Data Scientist), built an AI/ML Accelerator that cut modeling cycles ~75% and an ML suite proving a 4.1% registration uplift worth $9.6M in new annual revenue.

Brane (SPLUSM LLC) **Co-Founder & CEO** Orlando, FL · 2020 - Present

- Co-founded and lead Brane Mobility, a cloud-native B2B SaaS that turns raw mobility and urban data into multi-scenario, stakeholder-ready insights in minutes for city planners, public agencies, and enterprises. Built and led a ~25-person cross-functional org (GenAI and data scientists, data engineers, full-stack developers, UX/UI) and awarded $625K in grants.
- In 2026 launched Brane AI on the same platform: a data-center siting and approval-intelligence product that screens every census block group in a three-county corridor against 40 indicators (six gating), prices nine mitigation levers with benefit-cost and lifecycle math, and serves developers and county, utility, and water-district reviewers from one scoring object. Every on-screen number reproduces through the product's own API, enforced by a six-layer test gate; nine roadmap phases shipped.
- Architected the platform as two integrated systems: a data-management layer (a four-level dataset-to-KPI hierarchy with automated ETL, validation, a custom KPI formula builder, and user-data integration) and a five-step analysis workflow (project definition across 18 problem types, hotspot identification, multi-objective optimization, sensitivity analysis, and automated Word, PDF, and slide reporting). The AI Decision Engine pairs neural-network forecasting and genetic-algorithm optimization with an AWS Bedrock copilot; multi-city pilots showed ~18% operational efficiency gains.

UCF College of Engineering & Computer Science **Lead Data Scientist & Researcher** Orlando, FL · 2017 - 2020

- Delivered two production applications with end-to-end data pipelines used by the U.S. Department of Transportation for traffic and cost forecasting; secured competitive USDOT and NSF funding. Led applied-AI research in forecasting, optimization, and simulation, producing 20+ peer-reviewed publications.

Stratus International Contracting Co. **Senior Data Analyst** · 2011 - 2017

- Built a PMO optimization framework (hybrid genetic algorithm plus constraint-based simulation) and stood up the firm's first PMI-aligned PMO, cutting 21% of cost and 32% of time across major projects in three years.

SELECTED PRODUCTION AI SOLUTIONS (BUSINESS OUTCOME FIRST)

- Cut payer-policy and claims analysis time 90% with a 10% accuracy gain for a top-5 health payer, surfacing millions in recoverable overpayments. A 9-agent LangGraph pipeline generates 200+ Snowflake-ready SQL queries per policy; reused across payers.
- Cut invoice processing time 92% with a 5% accuracy gain across an airline, a manufacturer, and a large health system with a straight-through invoice agent on AWS Bedrock (Textract plus validation), routing only exceptions to a human.
- Surfaced spend savings across 14,765 suppliers and 47,489 parts for a Fortune 500 airline: a 10-agent hand-orchestrated pipeline plus a React command center delivered a 98% task-time reduction and a 20% accuracy gain.
- Spend intelligence platform: a browser-native spend-intelligence platform (React + DuckDB-WASM, zero external requests) over AP spend, 240K MRO parts lines, and payer vendor cost, with an agentic analyst that writes and runs real SQL and cites its figures; every insight is SQL-computed, never LLM-generated, and release is gated by 160+ browser checks driven by six skills.
- Skills-driven deck builder: eight authored skills take source files to a client-ready PowerPoint on the corporate template; deterministic ground truth traces every dollar, and a QC gate stops the build if an unhedged figure misses its metric by more than 2%. 80% less deck time, zero untraced numbers.
- Cut unit-cost claims analysis time 35% with a 4% accuracy gain for a state Medicaid program using a 7-agent LangGraph cognitive loop, taking a five-day analysis down to hours.
- Protected margin on thousands of daily financing deals for an equipment-finance lender. An 8-agent pricing engine (XGBoost risk scoring, reinforcement-learning optimization, SHAP explainability) is targeting a 15 to 20% margin improvement and 60% faster decisions, explainable for compliance, for a newly won $700K+ engagement.
- Made underwriting transparent and regulator-ready with an 11-agent LangGraph system that scores 11 risk dimensions under a Chief Underwriter orchestrator (XGBoost + SHAP, every decision explained). Open source.
- Built contract review that ships itself: a 7-agent clause-assessment engine rates every clause P/A/D/U from a tool-called rubric (criteria never hard-coded), with deterministic guards, an independent QC agent, and human routing for anything uncertain; 85% of assessments validated to auto-approve.
- Caught the fraud rings rules miss with a 3-agent LangGraph investigation (graph neural net plus LLM) over a 3D transaction graph. Open source.

EDUCATION & CERTIFICATIONS

Ph.D., Engineering, University of Central Florida. Dissertation: machine-learning algorithms for forecasting (ML / applied-AI methods).
M.S., Engineering, UCF. Executive Education, Business Administration, Harvard Business School.

Certifications: PMP · PMI-PBA · Certified SAFe Practitioner · Advanced Data Science Fellowship (The Data Incubator) · Claude Certified Architect, Anthropic (expected 2026) · AWS Solutions Architect Associate (expected 2026).

PUBLICATIONS, PATENTS & RECOGNITION

Patent & features: U.S. Patent (2025), GenAI contract-intelligence method (Doczy.ai) · featured on the AWS Architecture Blog · founder of Modern AI Academy, teaching how agentic AI is built and shipped.

Research: 20+ peer-reviewed publications, 460 citations, h-index 10, i10-index 10 (Google Scholar, Sept 2026), in IEEE Access, MDPI, and the ASCE Journal of Construction Engineering & Management. Most-cited: drivers and barriers to connected, automated, shared, and electric vehicles (IEEE Access, 139 citations). Reviewer for the Journal of Intelligent Transportation Systems and TRB.